Archita

Boston, MA | (857) 867-5859 | archita.l@northeastern.edu | LinkedIn | Github | Portfolio

EDUCATION

Northeastern University, Boston, MA **Sept 2025 - May 2027**

Master's in Artificial Intelligence

Relevant Coursework: Machine Learning, Linear Algebra, Natural Language Processing, Algorithms, MLOps

Guru Gobind Singh Indraprastha University | Delhi, India **Sept 2020 – Dec 2024**

Bachelor's in Computer Science

SKILLS

Languages & Databases: Python, C/C++, R, Snowflake, PostgreSQL, MySQL, Git/Github
Frameworks: TensorFlow, PyTorch, Keras, Scikit-Learn, Matplotlib, Seaborn, OpenCV, Transformers, Hugging Face
AI & ML: Deep Learning, Reinforcement Learning, CV, NLP, Predictive Analysis**,** Time Series Forecasting
Generative AI: LLMs (GPT), GANs, Prompt Engineering, RAG (Retrieval-Augmented Generation)
Tools: Jupyter Notebook, Google Colab, Power BI
Cloud: AWS (SageMaker, Lambda), Docker

EXPERIENCE

Enterprise Resource Planning Consultant | Technip Energies, India *Mar 2025 - Aug 2025*

- Developed an AI-driven ERP access validation and role recommendation workflow in ServiceNow using Python and SQL, combining SoD rules with similarity-based scoring on ticket creation to handle 100+ tickets per week and onboard 40+ users, reducing tickets needing manual review by 25%.
- Implemented human-in-the-loop approvals with confidence scoring and a threshold of 0.80 to route low-confidence cases for review, as a ServiceNow scheduled job for periodic access reviews and role cleanup, while standardizing role templates to enforce least privilege.

Machine Learning Intern | C-DAC (MeitY, Government of India), Delhi *Nov 2023 - May 2024*

- Built a real-time drowsiness detection pipeline on live webcam streams using OpenCV, Keras, and dlib, fine-tuning MobileNet on nearly 10K public and custom images with frame resizing for faster inference, reaching 98% accuracy and a 30% improvement in prediction.
- Implemented temporal rules using an (eyes closed >4s, yawns/min) threshold and yawn counts per minute, reducing false positives for yawns, and validated performance under glasses and low-light conditions using accuracy, precision, recall, F1, and confusion matrix.

PROJECTS

SmallHands: Compressing/Stress-Testing Vision-Language-Action Robot Policies

Link | PyTorch, LeRobot, Hugging Face, LIBERO (MuJoCo)

- Fine-tuned SmolVLA, a 450M-parameter model that maps camera images and text instructions to robot actions, reaching 65% closed-loop task success in simulation on LIBERO (100 episodes, 95% CI 55–74%).
- Showed in a controlled comparison that training only the action module matched full language-layer fine-tuning at lower compute cost, and that lower training loss did not predict higher task success.
- Exposed a language-grounding weakness: wrong instructions cut success to 6%, proving the policy uses language, yet simply rewording correct ones cut it to 0–26%, showing it relies on exact training phrasing rather than meaning.

Mint Sage (AI-Based Personal Finance Assistant)

Link | NLP, MiniLM, SVM/XGBoost, Ensemble Learning

- Developed three embedding strategies (TF-IDF → Universal Sentence Encoder → MiniLM/Sentence-BERT) to identify the strongest semantic representation for short, noisy transaction text, then built a GridSearchCV-tuned SVM + XGBoost ensemble with probability-weighted voting, reaching 90% accuracy across 20 expense categories.
- Built the end-to-end categorization pipeline-preprocessing, rare-class filtering, feature engineering - and shipped it as an integration-ready API consumed by the team's email-parsing, transaction-tracking, and forecasting modules; ran error analysis to diagnose confusion between semantically overlapping categories.

Sentiment Analyser

Link | BERT, Transformers, NLTK, Natural Language Processing (NLP)

- Developed a BERT sentiment classifier (Hugging Face Transformers) with NLTK-based preprocessing (cleaning, tokenization, stopword removal) on ~50K labeled Reddit and Twitter posts across positive/negative/neutral classes, reaching 85% accuracy.
- Generated topic-level sentiment tables and week-over-week trend reports using TF-IDF clustering in Jupyter, exporting CSV reports to flag topic-specific spikes in negative sentiment for feedback analysis.

PUBLICATIONS

- **"**Chatbots for Healthcare: A Literature Review". ICCTRDA| **Sep 2025 | Link**
- **"**Non-Invasive Health Monitoring and Fitness Scoring System: A Literature Review". IJHTM **| Nov 2024**